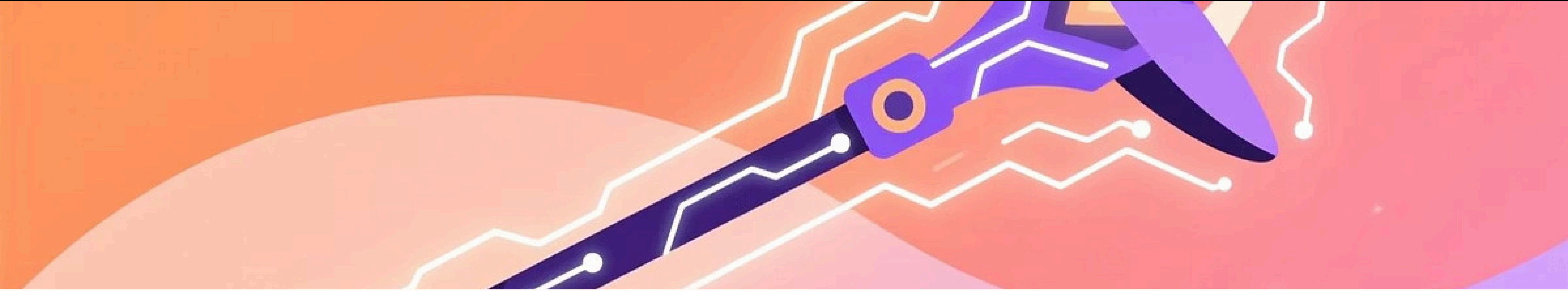
Gandalf's Staff: Navigation Reimagined

A pervasive computing project that transforms Tolkien's legendary artifact into a real-world navigation companion through modern sensors and AI-driven interaction.

T1-Group 5



Design Vision

Mythical Inspiration

Gandalf's staff symbolises light, guidance, hope, and protection in Tolkien's stories.

We captured that "magic" through technology by creating a smart walking staff that helps users explore safely and confidently.

Technical Reality

ESP32-E microcontroller, GPS module, magnetometer, button, and NeoPixel ring work together.

Real-time sensing, context-awareness, and adaptive response make it both practical aid and playful fantasy.

User Experience Flow

Input Phase

Press button to set exploration distance. Yellow LEDs confirm input.

Processing Phase

White glow while AI calculates destination using weather and location data.

Navigation Phase

Blue LEDs illuminate direction to destination. Dynamic updates as you move.

Arrival Phase

Yellow flash at 50m. Rainbow celebration at destination.

System Architecture

Hardware Core

- ESP32-E microcontroller
- NEO-6M GPS
- MMC5603 magnetometer
- NeoPixel ring
- 3.7V 1000mAh Li-ion battery

Software Intelligence

- Gemini API for context
- Weather API for adaptation
- Places API for coordinates
- Real-time sensor fusion
- Dynamic LED targeting

Physical Design

- Modular 3D-printed enclosure with 1.5 m wooden pole
- Designed for easy maintenance and durability outdoors
- Ergonomic, one-handed operation

5h	2.5m	<1s
Battery Life	**GPS Accuracy**	**Response Time**
Continuous operation with all systems active	Horizontal positioning precision (CEP)	Sensor update to LED feedback latency

Results & Future Vision

Proven Performance

GPS module delivered consistent tracking with 6-8 satellites and 2-3m accuracy. Navigation algorithms correctly computed bearing and distance. LED guidance responded dynamically to user orientation changes.

From Prototype to Product

This project demonstrates how mythical objects can be reinterpreted through pervasive computing, embedding intelligence seamlessly into familiar physical forms.

Future enhancements: voice feedback, machine learning for personalised suggestions, improved power efficiency, and commercial-grade durability.

> **Key Achievement:** Successfully transformed a legendary fantasy artifact into a functional context-aware navigation system that blends storytelling with precision engineering.